AQ7.4: Activity Questions 4 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
Consider a function $f: V \times V \rightarrow R$f : V × V → R where $V \subseteq R^2$V ⊆ R2 defined by $f(v, w) = 2v_1 w_1 + 5v_2 w_2$f(v, w) = 2v1w1 + 5v2w2, where $v = (v_1, v_2), w = (w_1, w_2)$v = (v1, v2), w = (w1, w2). Choose the set of correct options.

- [ ] ff satisfies the symmetry condition of the inner product.
- [ ] ff satisfies the bilinearity condition of the inner product.
- [ ] ff satisfies the positivity condition of the inner product.
- [ ] ff is an inner product.
- [ ] ff is not an inner product.

No, the answer is incorrect.
Score: 0
Accepted Answers:
ff satisfies the symmetry condition of the inner product.
ff satisfies the bilinearity condition of the inner product.
ff satisfies the positivity condition of the inner product.
ff is an inner product.

1 point
Consider a function $f: V \times V \rightarrow R$f : V × V → R where $V \subseteq R^2$V ⊆ R2 defined by $f(v, w) = v_1 w_1 - v_1 w_2 - v_2 w_1 + 4v_2 w_2$f(v, w) = v1w1 − v1w2 − v2w1 + 4v2w2, where $v = (v_1, v_2), w = (w_1, w_2)$v = (v1, v2), w = (w1, w2). Choose the set of correct options.

- [ ] ff satisfies the symmetry condition of the inner product.
- [ ] ff satisfies the bilinearity condition of the inner product.
- [ ] ff satisfies the positivity condition of the inner product.
- [ ] ff is an inner product.
- [ ] ff is not an inner product.

No, the answer is incorrect.
Score: 0
Accepted Answers:
ff satisfies the symmetry condition of the inner product.
ff satisfies the bilinearity condition of the inner product.
ff satisfies the positivity condition of the inner product.
ff is an inner product.

1 point

Consider two vectors $a = (0.4, 1.3, -2.2), b = (2, 3, -5)$a = (0.4, 1.3, −2.2), b = (2, 3, −5) in R^3 R3. Choose the set of correct options.

- [ ] The two vectors satisfy the triangle inequality given by $\| a + b \| \leq \| a \| + \| b \|$.‖a + b‖ ≤ ‖a‖ + ‖b‖ .
- [ ] The two vectors do not satisfy the triangle inequality.
- [ ] The two vectors satisfy the Cauchy-Schwarz inequality given by $| < a, b > | \leq \| a \| \| b \|$.| < a, b > | ≤ ‖a‖ ‖b‖ .
- [ ] The two vectors do not satisfy the Cauchy-Schwarz inequality.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The two vectors satisfy the triangle inequality given by $\| a + b \| \leq \| a \| + \| b \|$.‖a + b‖ ≤ ‖a‖ + ‖b‖ .

The two vectors satisfy the Cauchy-Schwarz inequality given by $| < a, b > | \leq \| a \| \| b \|$.| < a, b > | ≤ ‖a‖ ‖b‖ .

Level 2

1 point

Consider a function $f : V \times V \rightarrow R$f : V × V → R where $V \subseteq R^2$ V ⊆ R2 defined by $f(v, w) = v_1^2 w_1^2 + v_1 w_2^2 + v_2^2 w_1$ f(v, w) = v12w12 + v1w22 + v22w1, where $v = (v_1, v_2), w = (w_1, w_2)$v = (v1, v2), w = (w1, w2). Choose the set of correct options.

- [ ] f f satisfies the symmetry condition of the inner product.
- [ ] f f satisfies the positivity condition of the inner product.
- [ ] f f is an inner product.
- [ ] f f is not an inner product.

No, the answer is incorrect.

Score: 0

Accepted Answers:

f f satisfies the symmetry condition of the inner product.

f f is not an inner product.

1 point

Consider a vector $a = (a_1, b_1, c_1)$a = (a1, b1, c1) in R^3 R3. Which of these is (are) possible candidates for a norm?

- [ ] $\sqrt{a_1^2 + b_1^2 + c_1^2}$ a12 + b12 + c12
- [ ] $\sqrt{\quad}$
- [ ] $a_1 - b_1$ a1 − b1

$a_1 + b_1 + c_1$ a1 + b1 + c1

☐

$max\{a_1, b_1, c_1\}$max{a1, b1, c1}

☐

$min\{a_1, b_1, c_1\}$min{a1, b1, c1}

☐

max{$|a_1|, |b_1|, |c_1|$}max{|a1|, |b1|, |c1|}

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\sqrt{a_1^2 + b_1^2 + c_1^2}$ a12 + b12 + c12

$\sqrt{\quad}$

max{$|a_1|, |b_1|, |c_1|$}max{|a1|, |b1|, |c1|}

1 point

Choose the set of correct statements.

☐

$V = R^2$ V = R2 and the function $< ., . > : V \times V \longrightarrow R$< ., . >: V × V ⟶ R defined as $< (x_1, x_2), (y_1, y_2) >$ < (x1, x2), (y1, y2) > = $5x_1 y_1 + 8x_2 y_2 - 6x_1 y_2 - 6x_2 y_1$ 5x1y1 + 8x2y2 − 6x1y2 − 6x2y1 is an inner product on VV .

☐

$V = R^2$ V = R2 and the function $< ., . > : V \times V \longrightarrow R$< ., . >: V × V ⟶ R defined as $< (x_1, x_2), (y_1, y_2) >$ < (x1, x2), (y1, y2) > = $|x_1| + 2|y_2|$ |x1| + 2|y2| is an inner product on V V .

☐

$V = R^2$ V = R2 and the function $< ., . > : V \times V \longrightarrow R$< ., . >: V × V ⟶ R defined as $< (x_1, x_2), (y_1, y_2) >$ < (x1, x2), (y1, y2) > =$3x_1 y_1 + 5x_2 y_2$ 3x1y1 + 5x2y2 is an inner product on VV .

☐

$V = R^2$ V = R2 and the function $< ., . > : V \times V \longrightarrow R$< ., . >: V × V ⟶ R defined as $< (x_1, x_2), (y_1, y_2) >$ < (x1, x2), (y1, y2) > =$3x_1 y_1 - 5x_2 y_2$ 3x1y1 − 5x2y2 is an inner product on VV .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$V = R^2$ V = R2 and the function $< ., . > : V \times V \longrightarrow R$< ., . >: V × V ⟶ R defined as $< (x_1, x_2), (y_1, y_2) >$ < (x1, x2), (y1, y2) > = $5x_1 y_1 + 8x_2 y_2 - 6x_1 y_2 - 6x_2 y_1$ 5x1y1 + 8x2y2 − 6x1y2 − 6x2y1 is an inner product on VV .

$V = R^2$ V = R2 and the function $< ., . > : V \times V \longrightarrow R$< ., . >: V × V ⟶ R defined as $< (x_1, x_2), (y_1, y_2) >$ < (x1, x2), (y1, y2) > =$3x_1 y_1 + 5x_2 y_2$ 3x1y1 + 5x2y2 is an inner product on VV .

Check Answers

Your score is: 0/6